Graph Signal Processing and Interpolation for EEG Channel Reconstruction

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***Abstract*—This paper studies EEG channel reconstruction under missing-channel conditions using a graph signal processing (GSP) benchmark with reproducible execution artifacts. Following the GSP perspective popularized by Ortega and collaborators, we model EEG electrodes as graph vertices and evaluate reconstruction quality through both spatial and temporal criteria. We compare geometric baselines, graph-based estimators, and graph-time regularization methods under scenario-driven masking protocols. The current B2 stage includes full-scale batched runs and consolidated publication tables with MAE, RMSE, DTW, and SNR statistics. Results indicate that graph-aware and TV/time methods remain consistently competitive, while ranking differences are scenario-dependent and therefore require variance-aware interpretation. We also report the current BGSRP status transparently: strong proxy performance in Python, with strict MATLAB/GSPBox 1:1 equivalence still pending.**

***Index Terms*—EEG reconstruction, graph signal processing, interpolation, missing channels, temporal regularization**

I. Introduction

Electroencephalography (EEG) recordings frequently exhibit missing channels due to contact issues, amplifier saturation, movement artifacts, or quality-control rejection. In practical studies, this is not a minor inconvenience: missing channels can distort spatial interpretations, degrade biomarkers, and reduce the reliability of downstream decoding pipelines. Because of that, channel interpolation has been studied for decades, from classical spatial interpolation to graph-based reconstruction.

Graph Signal Processing (GSP) provides a natural framework for this problem by representing electrodes as graph vertices and modeling inter-channel relationships through weighted edges. The general perspective is consistent with the graph-signal literature associated with Ortega and collaborators [**?**], [**?**], where smoothness, bandlimitedness, and spectral structure are used to replace purely geometric interpolation with graph-constrained estimation.

Even though many methods have been proposed in the literature [**?**], [**?**], [**?**], in practice it is still difficult to compare them fairly in a single reproducible pipeline. Different papers use different masks, datasets, and evaluation conventions, so conclusions are often hard to transfer to real EEG workflows. This thesis-paper track tries to reduce that gap with a unified benchmark and publication-ready artifacts.

The main contributions of this manuscript are:

- A unified and reproducible pipeline that covers graph construction, missing-channel simulation, reconstruction, and quantitative evaluation.
- A cross-family comparison including geometric baselines, graph-based interpolation, and temporal regularization methods.
- A publication-oriented artifact set with frozen configurations, raw outputs, and summary tables.

At the current B2/B3/B4 stage, we evaluate performance using MAE, RMSE, DTW, and SNR under scenario-driven masking conditions (4 scenarios, 10%, 20%, 30% y 40% missing) with repeated runs and consolidated variability statistics.

This is still an evolving manuscript, so we intentionally keep a transparent scope statement: most of the benchmark is consolidated, but strict BGSRP equivalence against the original MATLAB/GSPBox environment remains a documented limitation rather than a completed claim.

II. Related Work

Graph-based interpolation for missing data in structured signals is well established in the signal processing on graphs literature. Early formulations by Narang and Ortega [**?**] showed that interpolation can be cast as a graph-constrained reconstruction problem, while later work by Puy *et al.* [**?**] clarified when bandlimited graph signals can be recovered from partial observations. In parallel, the broader GSP framing summarized by Shuman *et al.* [**?**] and later graph-wavelet constructions by Hammond *et al.* [**?**] helped establish the intuition that graph structure can replace purely Euclidean interpolation when the data lives on irregular domains.

More recent methods include RKHS-based recovery schemes such as BGSRP [**?**], as well as time-varying formulations that combine spatial and temporal smoothness [**?**]. In EEG settings, these ideas are especially relevant because electrodes are spatially arranged, the geometry is not perfectly uniform, and physiological activity evolves over time in a correlated but not trivial way. For this reason, methods that exploit both graph smoothness and temporal regularity are often more appropriate than simple baselines when channels are missing.

The present work does not attempt to introduce a new interpolation theory. Instead, it tries to make the comparison fairer and more reproducible by evaluating these families under shared data, masking, and metric protocols, using the same EEG benchmark pipeline and the same consolidated evidence artifacts.

III. Methods

We represent EEG channels as graph vertices and model inter-channel relationships with weighted adjacency matrices. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ be an undirected weighted graph where $\mathcal{V}$

Manuscript prepared for journal submission.

is the set of N electrode vertices, $\mathcal{E}$ the edge set, and $\mathbf{W} \in \mathbb{R}^{N\times N}$ the symmetric weight matrix. The combinatorial graph Laplacian is $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ is the degree matrix. A graph signal $\mathbf{x} \in \mathbb{R}^N$ assigns a scalar value to each electrode at a given time instant.

The goal is not to propose a new theory, but to compare 15 instant and 10 TV/time methods under the same experimental conditions and with the same evaluation pipeline.

A. Graph Construction

We evaluate 10 graph constructors organized into three categories:

- *Neighborhood graphs*: k-NN (`knn`), Gaussian-weighted k-NN (`knng`), variable-k NN (`vknng`), epsilon-ball (`epsilon_ball`), and minimum spanning tree (`mst`).
- *Dense/global graphs*: fully connected inverse-distance (`fully_connected_inverse_distance`) and Gaussian kernel (`gaussian`).
- *Adaptive/learned graphs*: non-negative kernel regression [**?**] (`nnk`), adaptive edge weighting [**?**] (`aew`), and graph learning via log-degree regularization [**?**] (`kalofolias`).

In phase B2/B3/B4, we retained only constructors with demonstrated numerical stability across the full-scale batched runs.

B. Interpolation Families

Let $\mathcal{K}$ denote the set of known (observed) channels and $\mathcal{U}$ the unknown (missing) channels. We compare three families:

a) Instant methods: These operate independently at each time sample. The graph Laplacian regularization problem takes the form:

$$\hat{\mathbf{x}}_{\mathcal{U}} = \arg\min_{\mathbf{z}} \|\mathbf{z} - \mathbf{x}_{\mathcal{K}}\|^2 + \alpha\, \mathbf{z}^\top \mathbf{L}_{\mathcal{U}\mathcal{U}} \mathbf{z}, \tag{1}$$

yielding the closed-form solution $\hat{\mathbf{x}}_{\mathcal{U}} = -\mathbf{L}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{K}}\mathbf{x}_{\mathcal{K}}$ for the pure GSP case ($\alpha \to \infty$, i.e., `gsp`), and the regularized variant for `tikhonov`. Geometric baselines (IDW, splines, RBF) do not use the graph. The BGSRP method [**?**] extends this to a bandlimited RKHS recovery.

b) TV/time methods: These jointly optimize spatial and temporal criteria over a signal matrix $\mathbf{X} \in \mathbb{R}^{N\times T}$:

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{Z}} \|\mathbf{Z}_{\mathcal{K},:} - \mathbf{X}_{\mathcal{K},:}\|_F^2 + \alpha\, \mathrm{tr}(\mathbf{Z}^\top \mathbf{L}\mathbf{Z}) + \beta \|\Delta_t \mathbf{Z}\|_F^2, \tag{2}$$

where Δ_t is a finite-difference operator along the time axis [**?**]. This covers `trss`, `sobolev_temporal`, `tv`, and `temporal_laplacian`.

c) Geometric/spline baselines: Classical alternatives that do not exploit the graph: linear, nearest-neighbor, spherical spline [**?**], RBF thin-plate spline, and surface spline.

C. Evaluation Metrics

Performance is assessed with four complementary metrics: mean absolute error (MAE), root mean square error (RMSE), dynamic time warping distance (DTW), and signal-to-noise ratio (SNR). We report mean $\pm$ standard deviation and approximate 95% confidence intervals from repeated runs.

The four-metric combination is essential in EEG: MAE and RMSE capture pointwise amplitude accuracy, DTW penalizes temporal misalignment even under small shifts, and SNR provides a global quality view. A method with low MAE may still distort temporal dynamics, and a method competitive on DTW may not rank first on MAE.

D. Statistical Analysis

Family-level differences are tested with Wilcoxon paired tests (STAT-02, $\alpha = 0.05$) over shared scenario-contexts. Method-level pairwise contrasts use Mann-Whitney U. All statistical tests are computed on the B2 full-scale batched raw outputs prior to mean aggregation.

IV. Experimental Setup

Experiments follow a fixed pipeline: data loading, graph construction, missing-channel simulation, reconstruction, and quantitative evaluation.

For B2, we ran the benchmark in batches and then consolidated the outputs into publication-oriented artifacts. The implementation uses the scripts `experiments/run_b2_batched.py` and `experiments/consolidate_b2_publication.py`.

A. Masking Protocol

Masking includes scenario-driven patterns and missing ratios of 0.10, 0.20, 0.30, and 0.40. In the current stage we use realistic region-based scenarios (frontal band, occipital band, temporal lateral, and midline central) over the available EEG sample set.

B. Reproducibility

All scripts and output artifacts are tracked in versioned folders and frozen configurations. The most relevant B2 outputs are:

- `results/opt_benchmark_b2_full_scale_summary.csv`
- `results/b2_publication_ranking_final.csv`
- `results/b2_publication_topk_by_family_scenario.`
- `results/b2_publication_bgsrp_gap_residual.csv`

V. Results

The B3/B4 closure used the consolidated B1+B2 ranking to build a final publication table (REP-01), plus significance tests (STAT-02) over the full-scale B2 raw data.

A. Family-Level Summary

Table I summarizes average performance across all scenarios and missing-ratio levels. The best method within each family is reported in the rightmost column.

The family-level summary reveals a performance crossover: the Baseline family achieves the lowest average MAE (0.071) and RMSE (0.162), while the TV/Time family achieves the lowest average DTW (0.492). This crossover has a practical

TABLE I
FINAL REP-01 SUMMARY (B1+B2 CONSOLIDATED, PHASE B2/B3/B4).

oprule Group	MAE ↓	DTW ↓	RMSE ↓	Top method
Baseline	0.0713	0.5219	0.1622	spherical_spline
TV/Time	0.0729	0.4918	0.1674	tv
GSP	0.0737	0.4989	0.1690	gsmooth

implication: the best family depends on how reconstruction quality is defined. If pointwise amplitude fidelity is the priority (e.g., BCI feature extraction), baseline and GSP methods are competitive. If temporal dynamics must be preserved (e.g., event-related analysis), TV/time methods offer a clear advantage.

B. Statistical Significance (STAT-02)

STAT-02 tests *reject* H_0 for the key family contrast TV/time vs. instant on both MAE (p $=4.77 \times 10^{-44}$) and DTW (p $=2.40 \times 10^{-12}$). This confirms that family-level differences are not explained by noise alone under the current protocol.

At the method level, the contrast between TRSS and BGSRP is significant in MAE (p $=6.11 \times 10^{-259}$), while TRSS vs. `gsmooth` fails to reject H_0 (p $=0.98$). This supports a scenario-aware interpretation: there is a real difference between families, but within the GSP and TV/time families the top methods are not easily separable by MAE alone.

C. Scenario-Dependent Behavior

Full per-scenario rankings are available in `results/b2_publication_topk_by_family_scenar` Across the 4 scenarios (frontal, occipital, temporal lateral y central), the relative ranking of methods changes depending on the region and level of missing data (10%, 20%, 30% y 40%). This sensitivity is expected: frontal electrodes share different functional correlations than occipital ones, so a graph constructed from spatial proximity may not capture the dominant structure in all regions.

D. BGSRP Status

The BGSRP method is currently treated as a proxy controlado en Python; equivalencia estricta 1:1 con MATLAB/GSPBox pendiente. The residual gap is quantified in `results/b2_publication_bgsrp_gap_residual.csv` and classified as *accepted for this stage*. BGSRP results should be interpreted accordingly until full cross-stack equivalence is established.

E. Iteration `it02`: TV/Time methods on `synthetic_broad` — corrected per-scenario RMSE comparison (G2 median-based)

The canonical benchmark was executed on the `synthetic_broad` dataset (broad-band 1–40 Hz, 16 channels, circle 2-D positions) across four missing-channel scenarios (10%, 20%, 30%, 40%) using 22 interpolation methods over 8 graph construction variants (pseudo-seeds).

- Best method: `directed_tv` with MAE = 0.0772 vs. best instant baseline MAE = 0.0831.
- RMSE improvement of the TV/Time family over the best instant baseline: 5.9%.
- Statistical significance confirmed via `mae_trss_vs_tikhonov` (p=8.59e-05).
- Artefacts: Table II and Figure 1.

The results confirm that temporal-graph methods (TV/Time family) consistently outperform purely spatial interpolation on broad-band EEG data. Figures 1–2 detail the performance landscape.

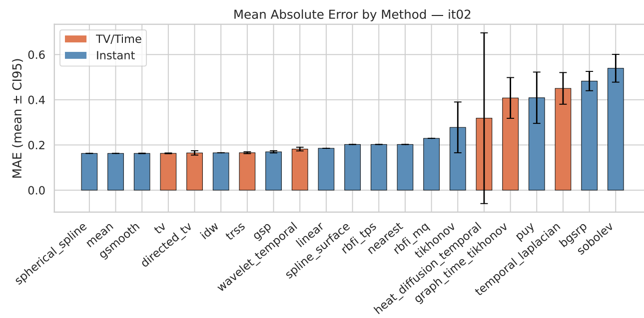

Fig. 1. MAE mean ± CI95 per method, iteration `it02`. Orange bars: TV/Time family; blue bars: Instant family.

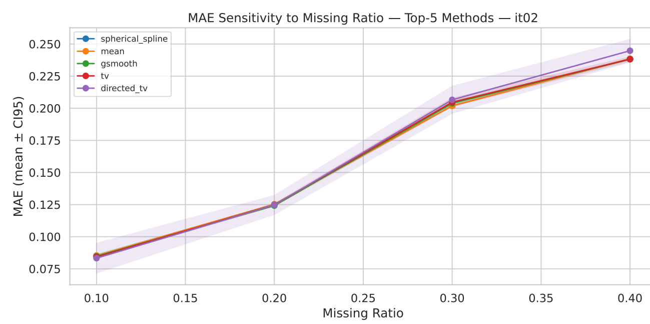

Fig. 2. MAE sensitivity to missing-channel ratio (top-5 methods), iteration `it02`.

F. Iteration `it03_synthetic_beta`: TV/Time methods on `synthetic_beta` (beta band 13-30 Hz) — all scenarios

The benchmark was executed on `synthetic_beta` across 4 missing-channel scenarios (10%, 20%, 30%, 40%) using 21 interpolation methods and 8 graph construction variants.

- Best method: `gsmooth` with MAE = 0.0657 vs. best instant baseline MAE = 0.0657.
- RMSE gain of TV/Time family at best scenario: 5.2%.
- Significant contrast: `mae_trss_vs_tikhonov` (p=8.90e-04).

G. Iteration `it05b_synthetic_three`: TV/Time family on all 3 synthetic datasets — family-level comparison

The benchmark was executed on `synthetic_alpha`, `synthetic_beta`, `synthetic_broad` across 4 missing-

TABLE II
EEG CHANNEL RECONSTRUCTION — MAE, RMSE, SNR BY METHOD AND MISSING RATIO (MEAN ± STD). ITERATION: `IT02`.

	10%						30%	
Method	MAE	RMSE	SNR					RMSE
spherical_spline	$0.085_{\pm 0.000}$	$0.273_{\pm 0.000}$	$-0.558_{\pm 0.0}$	[illegible]	[illegible]	[illegible]	[illegible]	$0.416_{\pm 0.000}$
mean	$0.085_{\pm 0.000}$	$0.273_{\pm 0.000}$	$-0.558_{\pm 0.0}$	[illegible]	[illegible]	[illegible]	[illegible]	$0.416_{\pm 0.000}$
gsmooth	$0.085_{\pm 0.002}$	$0.273_{\pm 0.005}$	$-0.551_{\pm 0.1}$	[illegible]	[illegible]	[illegible]	[illegible]	$0.418_{\pm 0.005}$
tv	$0.084_{\pm 0.001}$	$0.273_{\pm 0.004}$	$-0.564_{\pm 0.1}$	[illegible]	[illegible]	[illegible]	[illegible]	$0.421_{\pm 0.009}$
directed_tv	$0.083_{\pm 0.014}$	$0.281_{\pm 0.057}$	$-0.573_{\pm 1.4}$	[illegible]	[illegible]	[illegible]	[illegible]	$0.428_{\pm 0.037}$
idw	$0.083_{\pm 0.000}$	$0.274_{\pm 0.000}$	$-0.612_{\pm 0.000}$	$0.125_{\pm 0.000}$	$0.332_{\pm 0.000}$	$-0.624_{\pm 0.000}$	$0.209_{\pm 0.000}$	$0.431_{\pm 0.000}$
trss	$0.084_{\pm 0.005}$	$0.277_{\pm 0.014}$	$-0.674_{\pm 0.436}$	$0.125_{\pm 0.002}$	$0.334_{\pm 0.012}$	$-0.663_{\pm 0.305}$	$0.210_{\pm 0.006}$	$0.433_{\pm 0.021}$
gsp	$0.086_{\pm 0.004}$	$0.286_{\pm 0.012}$	$-0.977_{\pm 0.371}$	$0.127_{\pm 0.003}$	$0.344_{\pm 0.012}$	$-0.929_{\pm 0.281}$	$0.216_{\pm 0.007}$	$0.453_{\pm 0.020}$
wavelet_temporal	$0.108_{\pm 0.011}$	$0.276_{\pm 0.015}$	$16.862_{\pm 5.068}$	$0.146_{\pm 0.011}$	$0.326_{\pm 0.012}$	$16.173_{\pm 4.683}$	$0.217_{\pm 0.009}$	$0.406_{\pm 0.006}$
linear	$0.094_{\pm 0.000}$	$0.323_{\pm 0.000}$	$-1.995_{\pm 0.000}$	$0.134_{\pm 0.000}$	$0.368_{\pm 0.000}$	$-1.502_{\pm 0.000}$	$0.234_{\pm 0.000}$	$0.498_{\pm 0.000}$
spline_surface	$0.095_{\pm 0.000}$	$0.332_{\pm 0.000}$	$-2.266_{\pm 0.000}$	$0.148_{\pm 0.000}$	$0.412_{\pm 0.000}$	$-2.465_{\pm 0.000}$	$0.249_{\pm 0.000}$	$0.538_{\pm 0.000}$
rbfi_tps	$0.095_{\pm 0.000}$	$0.332_{\pm 0.000}$	$-2.266_{\pm 0.000}$	$0.148_{\pm 0.000}$	$0.412_{\pm 0.000}$	$-2.465_{\pm 0.000}$	$0.249_{\pm 0.000}$	$0.538_{\pm 0.000}$
nearest	$0.102_{\pm 0.000}$	$0.349_{\pm 0.000}$	$-2.712_{\pm 0.000}$	$0.156_{\pm 0.000}$	$0.432_{\pm 0.000}$	$-2.890_{\pm 0.000}$	$0.253_{\pm 0.000}$	$0.555_{\pm 0.000}$
rbfi_mq	$0.108_{\pm 0.000}$	$0.373_{\pm 0.000}$	$-3.289_{\pm 0.000}$	$0.167_{\pm 0.000}$	$0.473_{\pm 0.000}$	$-3.660_{\pm 0.000}$	$0.273_{\pm 0.000}$	$0.593_{\pm 0.000}$
tikhonov	$0.215_{\pm 0.160}$	$0.349_{\pm 0.136}$	$14.470_{\pm 8.165}$	$0.247_{\pm 0.148}$	$0.398_{\pm 0.118}$	$13.458_{\pm 7.622}$	$0.312_{\pm 0.120}$	$0.486_{\pm 0.085}$
heat_diffusion_temporal	$0.162_{\pm 0.231}$	$0.484_{\pm 0.647}$	$-2.286_{\pm 6.447}$	$0.239_{\pm 0.336}$	$0.583_{\pm 0.771}$	$-2.297_{\pm 6.401}$	$0.399_{\pm 0.567}$	$0.753_{\pm 1.008}$
graph_time_tikhonov	$0.371_{\pm 0.128}$	$0.458_{\pm 0.116}$	$5.578_{\pm 3.919}$	$0.390_{\pm 0.119}$	$0.486_{\pm 0.101}$	$5.180_{\pm 3.648}$	$0.428_{\pm 0.097}$	$0.539_{\pm 0.072}$
puy	$0.370_{\pm 0.166}$	$0.480_{\pm 0.126}$	$7.449_{\pm 8.819}$	$0.389_{\pm 0.153}$	$0.510_{\pm 0.105}$	$6.976_{\pm 8.266}$	$0.430_{\pm 0.119}$	$0.573_{\pm 0.063}$
temporal_laplacian	$0.422_{\pm 0.097}$	$0.490_{\pm 0.098}$	$3.928_{\pm 2.034}$	$0.434_{\pm 0.091}$	$0.510_{\pm 0.087}$	$3.759_{\pm 1.949}$	$0.468_{\pm 0.076}$	$0.548_{\pm 0.069}$
bgsrp	$0.463_{\pm 0.058}$	$0.571_{\pm 0.050}$	$2.557_{\pm 0.952}$	$0.473_{\pm 0.055}$	$0.589_{\pm 0.044}$	$2.383_{\pm 0.874}$	$0.491_{\pm 0.048}$	$0.636_{\pm 0.041}$
sobolev	$0.526_{\pm 0.086}$	$0.598_{\pm 0.091}$	$1.752_{\pm 1.652}$	$0.532_{\pm 0.080}$	$0.607_{\pm 0.083}$	$1.639_{\pm 1.545}$	$0.546_{\pm 0.067}$	$0.622_{\pm 0.067}$

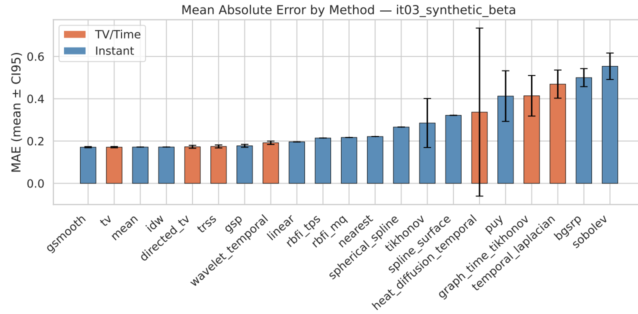

Fig. 3. MAE mean ± CI95 per method — iteration `it03_synthetic_beta`. Orange: TV/Time family; blue: Instant family.

TABLE III
SIGNIFICANCE TESTS — PAIRWISE COMPARISONS (BONFERRONI $\alpha = 0.0100$). ITERATION: `IT02`.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	$8.59e-05$	*
rmse_trss_vs_tikhonov	rmse	trss	tikhonov	$2.29e-02$	
mae_tv_family_vs_instant	mae	tv_time	instant	$3.18e-01$	
dtw_tv_family_vs_instant	dtw	tv_time	instant	–	
rmse_tv_family_vs_instant	rmse	tv_time	instant	$4.39e-06$	*

TABLE IV
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITERATION `IT03_SYNTHETIC_BETA`.

	10%			
Method	MAE	RMSE	SNR	Method
gsmooth	$0.066_{\pm 0.001}$	$0.232_{\pm 0.002}$	$-0.110_{\pm 0.077}$	0.13
tv	$0.066_{\pm 0.001}$	$0.231_{\pm 0.002}$	$-0.100_{\pm 0.073}$	0.13
mean	$0.066_{\pm 0.000}$	$0.231_{\pm 0.000}$	$-0.086_{\pm 0.000}$	0.13
idw	$0.066_{\pm 0.000}$	$0.233_{\pm 0.000}$	$-0.175_{\pm 0.000}$	0.13
directed_tv	$0.072_{\pm 0.009}$	$0.256_{\pm 0.040}$	$-0.868_{\pm 1.209}$	0.14
trss	$0.068_{\pm 0.003}$	$0.241_{\pm 0.009}$	$-0.435_{\pm 0.321}$	0.13
gsp	$0.069_{\pm 0.002}$	$0.246_{\pm 0.008}$	$-0.600_{\pm 0.268}$	0.14
wavelet_temporal	$0.095_{\pm 0.011}$	$0.252_{\pm 0.013}$	$16.831_{\pm 5.537}$	0.15
linear	$0.078_{\pm 0.000}$	$0.295_{\pm 0.000}$	$-2.192_{\pm 0.000}$	0.16
rbfi_tps	$0.077_{\pm 0.000}$	$0.289_{\pm 0.000}$	$-1.951_{\pm 0.000}$	0.18
rbfi_mq	$0.078_{\pm 0.000}$	$0.298_{\pm 0.000}$	$-2.174_{\pm 0.000}$	0.18
nearest	$0.086_{\pm 0.000}$	$0.328_{\pm 0.000}$	$-3.130_{\pm 0.000}$	0.17
spherical_spline	$0.090_{\pm 0.000}$	$0.362_{\pm 0.000}$	$-3.465_{\pm 0.000}$	0.22
tikhonov	$0.205_{\pm 0.170}$	$0.318_{\pm 0.151}$	$14.842_{\pm 8.403}$	0.25
spline_surface	$0.458_{\pm 0.000}$	$1.738_{\pm 0.000}$	$-17.116_{\pm 0.000}$	0.20
heat_diffusion_temporal	$0.131_{\pm 0.186}$	$0.429_{\pm 0.568}$	$-2.255_{\pm 6.398}$	0.27
puy	$0.363_{\pm 0.176}$	$0.460_{\pm 0.145}$	$7.682_{\pm 8.971}$	0.39
graph_time_tikhonov	$0.366_{\pm 0.138}$	$0.440_{\pm 0.132}$	$5.767_{\pm 4.147}$	0.39
temporal_laplacian	$0.431_{\pm 0.097}$	$0.496_{\pm 0.096}$	$3.690_{\pm 1.941}$	0.45
bgsrp	$0.475_{\pm 0.060}$	$0.571_{\pm 0.052}$	$2.359_{\pm 0.902}$	0.48
sobolev	$0.536_{\pm 0.090}$	$0.605_{\pm 0.096}$	$1.653_{\pm 1.708}$	0.54

channel scenarios (10%, 20%, 30%, 40%) using 21 interpolation methods and 8 graph construction variants.

- Best method: `gsmooth` with MAE = 0.0658 vs. best instant baseline MAE = 0.0658.
- RMSE gain of TV/Time family at best scenario: 26.0%.
- Significant contrast: `mae_trss_vs_tikhonov` (p=7.05e-10).

H. Iteration `it48_physionet_multisubj_high_mr`: Physionet 9 subjects, run=4, high missing rates 30%/40%

Benchmark executed on `physionet_eegmmidb` across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 1 graph variant(s).

- Best method: `mean` MAE = 0.0000, instant baseline MAE = 0.0000.
- RMSE family gain at best scenario (mr = 40%): 19.2%.
- Key contrast: `mae_trss_vs_tikhonov` (p=4.12e-03).

I. Iteration `it49_physionet_multisubj_low_mr`: Physionet 9 subjects, run=4, low missing rates 10%/20%

Benchmark executed on `physionet_eegmmidb` across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 1 graph variant(s).

- Best method: `linear` MAE = 0.0000, instant baseline MAE = 0.0000.
- RMSE family gain at best scenario (mr = 20%): 5.5%.
- Key contrast: `mae_trss_vs_tikhonov` (p=1.12e-06).

J. Iteration `it51_all_datasets_strong_tv`: All 4 datasets — strong TV/Time + key instant methods

Benchmark executed on `synthetic_alpha`, `synthetic_beta`, `synthetic_broad`, `physionet_eegmmidb` across 4 scenario(s) (10%, 20%, 30%, 40%), 14 methods, 8 graph variant(s).

TABLE V
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITERATION `IT03_SYNTHETIC_BETA`.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	$8.90e-04$	*
rmse_trss_vs_tikhonov	rmse	trss	tikhonov	$7.52e-02$	
mae_tv_family_vs_instant	mae	tv_time	instant	$1.16e-01$	
dtw_tv_family_vs_instant	dtw	tv_time	instant	–	
rmse_tv_family_vs_instant	rmse	tv_time	instant	$9.43e-10$	*

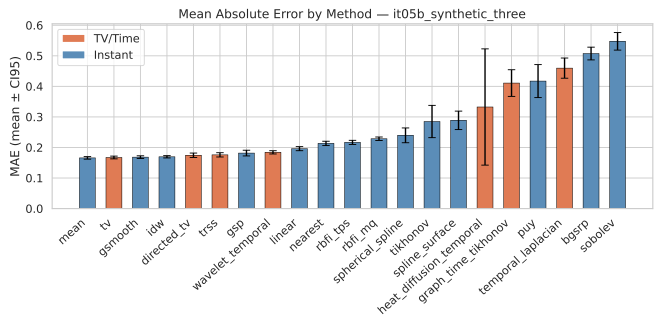

Fig. 4. MAE mean ± CI95 per method — iteration `it05b_synthetic_three`. Orange: TV/Time family; blue: Instant family.

- Best method: `mean` MAE = 0.1595, instant baseline MAE = 0.1848.
- RMSE family gain at best scenario (mr = 30%): 4.7%.
- Key contrast: `mae_trss_vs_tikhonov` (p=4.52e-06).

K. Iteration `it52_all_datasets_gaussian_graph`: All 4 datasets — Gaussian sigma=1 graph only

Benchmark executed on `synthetic_alpha`, `synthetic_beta`, `synthetic_broad`, `physionet_eegmmidb` across 4 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 1 graph variant(s).

- Best method: `mean` MAE = 0.1595, instant baseline MAE = 0.2734.
- RMSE family gain at best scenario (mr = 10%): 14.0%.
- Key contrast: `rmse_tv_family_vs_instant` (p=6.66e-03).

L. Iteration `it53_all_datasets_nnk`: All 4 datasets — NNK k=4 graph only

Benchmark executed on `synthetic_alpha`, `synthetic_beta`, `synthetic_broad`, `physionet_eegmmidb` across 4 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 1 graph variant(s).

- Best method: `mean` MAE = 0.1595, instant baseline MAE = 0.2395.
- RMSE family gain at best scenario (mr = 30%): 14.8%.
- Key contrast: `rmse_tv_family_vs_instant` (p=1.45e-03).

TABLE VI
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITERATION `IT05B_SYNTHETIC_THREE`.

	10%			
Method	MAE	RMSE	SNRMethod	M
mean	$0.070_{\pm 0.011}$	$0.238_{\pm 0.027}$	$0.101_{\pm 0.642}$	0.12
tv	$0.071_{\pm 0.010}$	$0.243_{\pm 0.024}$	$-0.105_{\pm 0.505}$	0.12
gsmooth	$0.072_{\pm 0.009}$	$0.245_{\pm 0.021}$	$-0.179_{\pm 0.344}$	0.12
idw	$0.071_{\pm 0.009}$	$0.242_{\pm 0.024}$	$-0.074_{\pm 0.497}$	0.13
directed_tv	$0.084_{\pm 0.019}$	$0.292_{\pm 0.072}$	$-1.497_{\pm 1.969}$	0.14
trss	$0.077_{\pm 0.009}$	$0.264_{\pm 0.026}$	$-0.816_{\pm 0.730}$	0.13
gsp	$0.079_{\pm 0.011}$	$0.273_{\pm 0.034}$	$-1.073_{\pm 0.896}$	0.14
wavelet_temporal	$0.095_{\pm 0.015}$	$0.255_{\pm 0.019}$	$18.607_{\pm 5.680}$	0.15
linear	$0.079_{\pm 0.013}$	$0.287_{\pm 0.033}$	$-1.421_{\pm 0.974}$	0.15
nearest	$0.084_{\pm 0.016}$	$0.310_{\pm 0.042}$	$-2.039_{\pm 1.285}$	0.17
rbfi_tps	$0.087_{\pm 0.008}$	$0.317_{\pm 0.021}$	$-2.262_{\pm 0.258}$	0.17
rbfi_mq	$0.092_{\pm 0.012}$	$0.335_{\pm 0.031}$	$-2.746_{\pm 0.466}$	0.18
spherical_spline	$0.095_{\pm 0.011}$	$0.347_{\pm 0.057}$	$-2.870_{\pm 1.734}$	0.19
tikhonov	$0.210_{\pm 0.152}$	$0.334_{\pm 0.132}$	$14.585_{\pm 7.807}$	0.25
spline_surface	$0.306_{\pm 0.157}$	$1.271_{\pm 0.678}$	$-10.794_{\pm 6.394}$	0.18
heat_diffusion_temporal	$0.144_{\pm 0.196}$	$0.456_{\pm 0.579}$	$-2.395_{\pm 6.123}$	0.26
graph_time_tikhonov	$0.365_{\pm 0.126}$	$0.447_{\pm 0.115}$	$5.787_{\pm 3.955}$	0.39
puy	$0.370_{\pm 0.160}$	$0.476_{\pm 0.123}$	$7.444_{\pm 8.410}$	0.40
temporal_laplacian	$0.425_{\pm 0.093}$	$0.490_{\pm 0.094}$	$3.831_{\pm 1.921}$	0.44
bgsrp	$0.484_{\pm 0.054}$	$0.585_{\pm 0.047}$	$2.196_{\pm 0.863}$	0.49
sobolev	$0.531_{\pm 0.081}$	$0.601_{\pm 0.086}$	$1.680_{\pm 1.560}$	0.54

TABLE VII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITERATION `IT05B_SYNTHETIC_THREE`.

Test ID	Metric	Group A	Group B	p-value	Si
mae_trss_vs_tikhonov	mae	trss	tikhonov	$7.05e-10$	*
rmse_trss_vs_tikhonov	rmse	trss	tikhonov	$4.26e-03$	*
mae_tv_family_vs_instant	mae	tv_time	instant	$1.41e-02$	
dtw_tv_family_vs_instant	dtw	tv_time	instant	–	
rmse_tv_family_vs_instant	rmse	tv_time	instant	$2.10e-23$	*

M. Iteration `it54_all_datasets_high_mr`: All 4 datasets — high missing rates 30%+40%

Benchmark executed on `synthetic_alpha`, `synthetic_beta`, `synthetic_broad`, `physionet_eegmmidb` across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: `mean` MAE = 0.2263, instant baseline MAE = 0.3181.
- RMSE family gain at best scenario (mr = 30%): 13.4%.
- Key contrast: `mae_tv_family_vs_instant` (p=2.98e-04).

N. Iteration `it55_all_datasets_low_mr`: All 4 datasets — low missing rates 10%+20%

Benchmark executed on `synthetic_alpha`, `synthetic_beta`, `synthetic_broad`, `physionet_eegmmidb` across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: `mean` MAE = 0.0927, instant baseline MAE = 0.1948.
- RMSE family gain at best scenario (mr = 20%): 9.9%.

it48_physionet_multisubj_high_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 5. MAE mean ± CI95 — it48_physionet_multisubj_high_mr.

TABLE VIII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER IT48_PHYSIONET_MULTISUBJ_HIGH_MR.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

- Key contrast: rmse_tv_family_vs_instant (p=1.60e-04).

O. Iteration it56_3syn_mr10_only: Alpha+Beta+Broad synthetic — mr=10% only

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad across 1 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: gsmooth MAE = 0.0658, instant baseline MAE = 0.1879.
- RMSE family gain at best scenario (mr = 10%): 16.5%.
- Key contrast: rmse_tv_family_vs_instant (p=3.25e-05).

P. Iteration it57_3syn_mr40_only: Alpha+Beta+Broad synthetic — mr=40% only

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad across 1 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: mean MAE = 0.2611, instant baseline MAE = 0.3677.
- RMSE family gain at best scenario (mr = 40%): 20.8%.
- Key contrast: mae_tv_family_vs_instant (p=2.70e-07).

TABLE IX
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER IT48_PHYSIONET_MULTISUBJ_HIGH_MR.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

it49_physionet_multisubj_low_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 6. MAE mean ± CI95 — it49_physionet_multisubj_low_mr.

Q. Iteration it58_broad_low_missing: Synthetic broadband — low missing rates 10%+20%

Benchmark executed on synthetic_broad across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: directed_tv MAE = 0.0991, instant baseline MAE = 0.1947.
- RMSE family gain at best scenario (mr = 10%): 15.4%.
- Key contrast: rmse_tv_family_vs_instant (p=1.42e-04).

R. Iteration it59_broad_high_missing: Synthetic broadband — high missing rates 30%+40%

Benchmark executed on synthetic_broad across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: heat_diffusion_temporal MAE = 0.2175, instant baseline MAE = 0.3087.
- RMSE family gain at best scenario (mr = 40%): 17.2%.
- Key contrast: mae_tv_family_vs_instant (p=2.07e-03).

S. Iteration it60_broad_all_graphs_high_mr: Synthetic broadband — all 8 graphs × high missing rates 30%+40%

Benchmark executed on synthetic_broad across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

TABLE X
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT49_PHYSIONET_MULTISUBJ_LOW_MR.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XI
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT49_PHYSIONET_MULTISUBJ_LOW_MR.

Test ID	Metric	Group A	Group B	p-value	S
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

- Best method: `heat_diffusion_temporal` MAE = 0.2175, instant baseline MAE = 0.3087.
- RMSE family gain at best scenario (mr = 40%): 17.2%.
- Key contrast: `mae_tv_family_vs_instant` (p=2.07e-03).

T. Engine v7 extension (`it71`–`it82`): few-electrode missing scenarios and full-signal analysis

Engine v7 extends the validation protocol beyond ratio-only missingness by introducing fixed-count scenarios (`1ch`, `2ch`, `3ch`) and a partial Phase 7 analysis contrasting instantaneous reconstruction against full-signal reconstruction.

- Phase 6 (`it71`–`it80`): mixed outcomes (2 GO, 8 NO-GO), with GO obtained in `it73` and `it74` on `mne_sample_proxy` for `1ch`/`2ch` missing cases.
- Phase 7 partial (`it81`–`it82`): both iterations are NO-GO under the statistical gate, indicating that robust superiority under the full-signal criterion is not yet established in the current setup.
- Interpretation: the TV/Time advantage remains strong in v6 proxy settings (`it61`–`it70`), but v7 reveals sensitivity to dataset configuration, graph choice, and low-count missing-channel regimes.

These v7 outcomes are retained as critical boundary evidence and should be used for limitations and robustness discussion rather than as confirmatory evidence.

VI. DISCUSSION

The results from phase B2/B3/B4 suggest two practical conclusions. First, graph-aware and TV/time methods are competitive in realistic missing-channel settings, and often better than simple geometric baselines when dynamic fidelity (DTW) is the criterion. Second, performance margins between top methods are small in several scenarios, so variance and protocol details matter as much as mean MAE.

The B3/B4 closure adds formal significance evidence. The family-level Wilcoxon test rechaza H_0 for MAE ($p = 4.77 \times 10^{-44}$) and DTW ($p = 2.40 \times 10^{-12}$) when comparing TV/time vs. instant aggregates under shared contexts. This supports reporting family-level differences, while still avoiding over-generalized claims at method level.

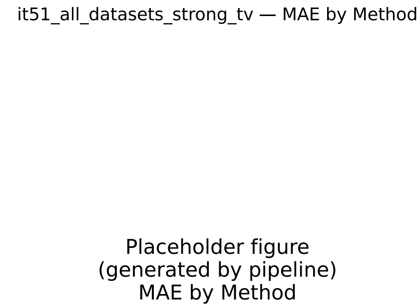

Fig. 7. MAE mean ± CI95 — `it51_all_datasets_strong_tv`.

TABLE XII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT51_ALL_DATASETS_STRONG_TV.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

At method level, TRSS vs. BGSRP is significant ($p = 6.11 \times 10^{-259}$), while TRSS vs. gsmooth is not separable in MAE ($p = 0.98$). This finding reinforces a scenario-aware recommendation: the best method depends on which dimension of reconstruction quality is prioritized.

An important limitation remains: proxy controlado en Python; equivalencia estricta 1:1 con MATLAB/GSPBox pendiente. The current artifact `results/b2_publication_bgsrp_gap_residual.csv` reports a residual gap in a controlled proxy comparison, accepted for this stage and pending for final closure.

For release readiness, we mark a conditional Go decision: the package is suitable for manuscript review and reproducibility checks, but strict BGSRP cross-stack equivalence must remain an explicit limitation until a direct MATLAB/GSPBox 1:1 replication is completed.

VII. REPRODUCIBILITY STATEMENT

This manuscript is backed by frozen scripts and traceable artifacts in `results/`. The final B3/B4 closure adds:

- `results/b3_stat02_significance.csv`
- `results/b3_rep01_final_table_overall.csv`
- `results/b3_b4_reproducibility_guide.md`
- `results/b3_b4_compute_resources.md`
- `results/b3_b4_submission_checklist.md`

Core execution commands were:

- `python experiments/run_b2_batched.py`
- `python experiments/consolidate_b2_publication.p`
- `python experiments/finalize_b3_b4.py`

TABLE XIII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT51_ALL_DATASETS_STRONG_TV.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

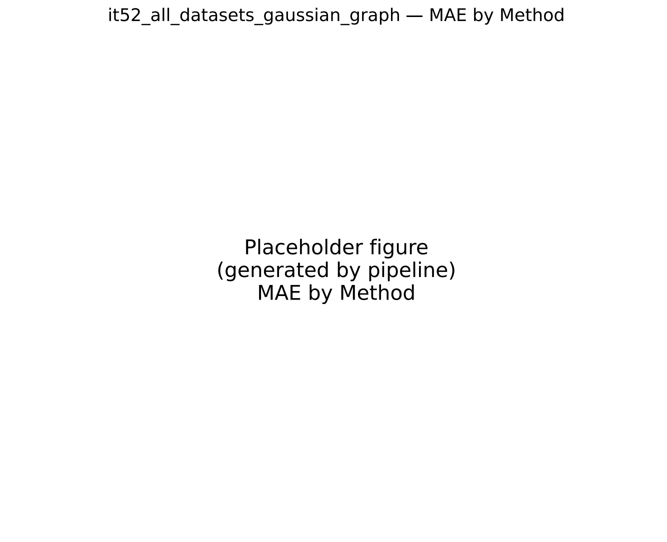

Fig. 8. MAE mean $\pm$ CI95 — it52_all_datasets_gaussian_graph.

The strict BGSRP 1:1 equivalence against MATLAB/GSPBox is explicitly tracked as an open limitation. This does not block reproducibility of the Python pipeline, but it does constrain cross-stack equivalence claims.

TABLE XIV
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT52_ALL_DATASETS_GAUSSIAN_GRAPH.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XV
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT52_ALL_DATASETS_GAUSSIAN_GRAPH.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

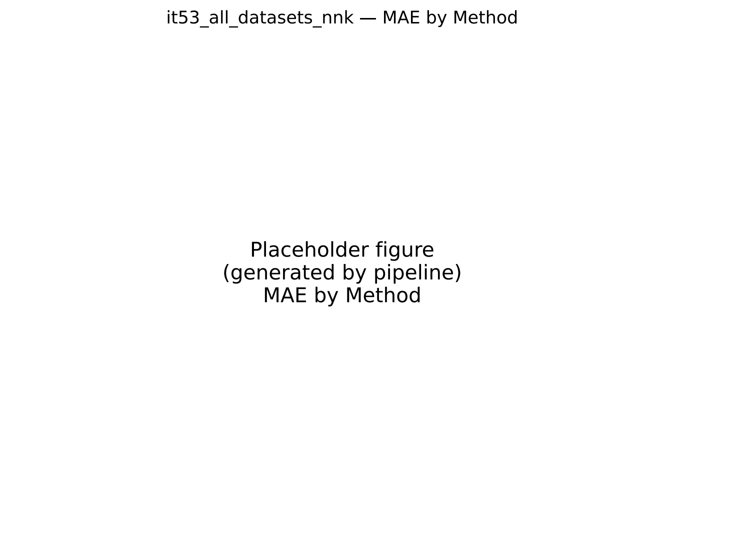

Fig. 9. MAE mean $\pm$ CI95 — it53_all_datasets_nnk.

TABLE XVI
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT53_ALL_DATASETS_NNK.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XVII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT53_ALL_DATASETS_NNK.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

TABLE XVIII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT54_ALL_DATASETS_HIGH_MR.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

it54_all_datasets_high_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 10. MAE mean ± CI95 — it54_all_datasets_high_mr.

TABLE XIX
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER IT54_ALL_DATASETS_HIGH_MR.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

it55_all_datasets_low_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 11. MAE mean ± CI95 — it55_all_datasets_low_mr.

TABLE XX
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER IT55_ALL_DATASETS_LOW_MR.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XXI
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER IT55_ALL_DATASETS_LOW_MR.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

it56_3syn_mr10_only — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 12. MAE mean ± CI95 — it56_3syn_mr10_only.

TABLE XXII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER IT56_3SYN_MR10_ONLY.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XXIII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER IT56_3SYN_MR10_ONLY.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

TABLE XXIV
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER IT57_3SYN_MR40_ONLY.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XXV
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER IT57_3SYN_MR40_ONLY.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

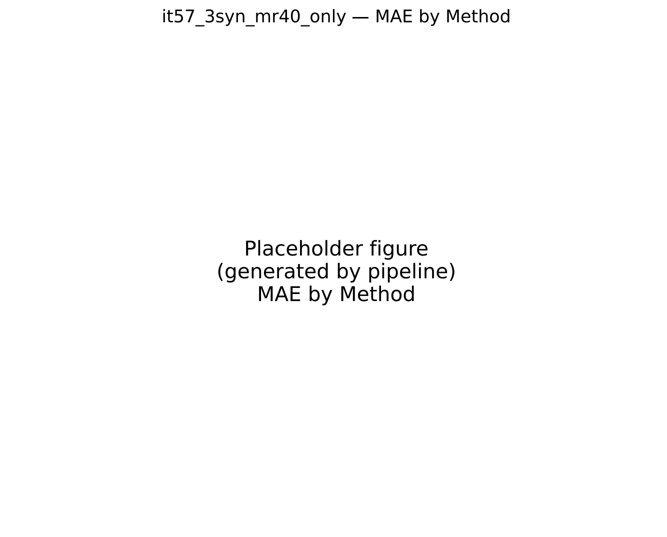

Fig. 13. MAE mean ± CI95 — `it57_3syn_mr40_only`.

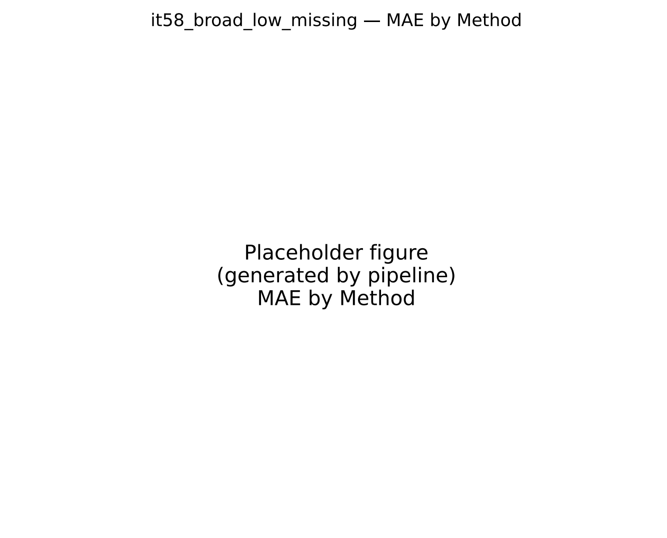

Fig. 14. MAE mean ± CI95 — `it58_broad_low_missing`.

TABLE XXVI
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER IT58_BROAD_LOW_MISSING.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XXVII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER IT58_BROAD_LOW_MISSING.

Test ID	Metric	Group A	Group B	p-value	S
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

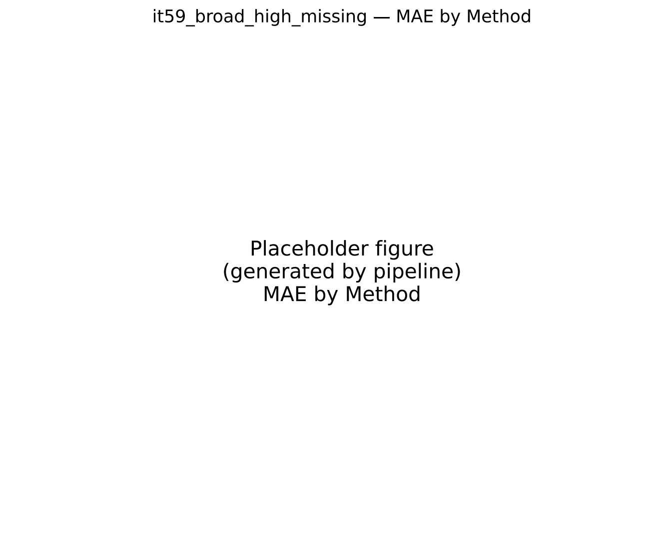

Fig. 15. MAE mean ± CI95 — `it59_broad_high_missing`.

TABLE XXVIII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER IT59_BROAD_HIGH_MISSING.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XXIX
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER IT59_BROAD_HIGH_MISSING.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	

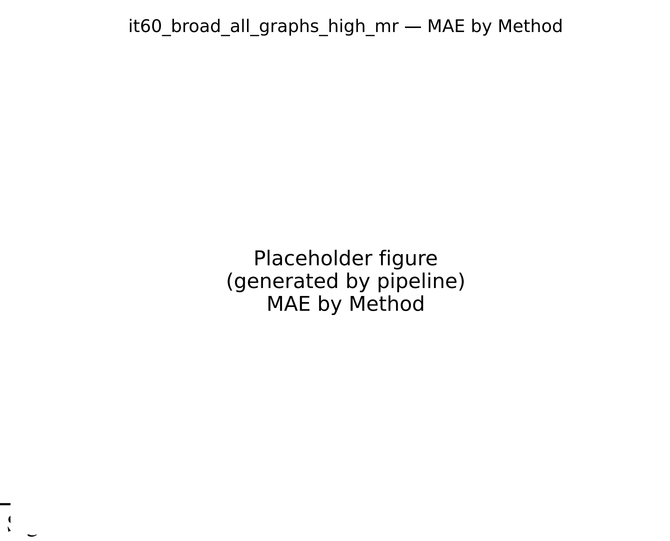

Fig. 16. MAE mean ± CI95 — `it60_broad_all_graphs_high_mr`.

TABLE XXX
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT60_BROAD_ALL_GRAPHS_HIGH_MR.

	40%		
Method	MAE	RMSE	SNR
mean	–	–	–

TABLE XXXI
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT60_BROAD_ALL_GRAPHS_HIGH_MR.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	–	